overview

Deployment Overview — Helix OTA 1.0.0-MVP

Field	Value
Revision	2
Created	2026-06-07
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Status	active
Status summary	Deployment topology for the Helix OTA 1.0.0-MVP control plane. Defines the containerized service set (ota-server, postgres, minio, reverse proxy, dashboard), the container substrate (`vasic-digital/containers` submodule), secrets handling (security brick + container secrets), and the four-layer testing gates for deployment artifacts. Honors the LOCKED stack (Go + Gin + Brotli + HTTP/3→HTTP/2, REST-primary; PostgreSQL + MinIO) and the LOCKED strategy (native Android A/B + custom Go modular-monolith control plane per ADR-0003).
Issues	HelixConstitution clause numbers are UNVERIFIED (carried from corpus convention). Exact public surface of the `containers`, `security`, `config`, and `observability` catalogue submodules has not been inspected; “satisfies” claims are UNVERIFIED where so marked. Concrete resource sizing (CPU/RAM/replica counts) and load-derived scale thresholds are UNVERIFIED — to be set from MVP load tests (ADR-0003 §3.2, §7). Reverse-proxy and dashboard images are placeholders pending build-pipeline confirmation.

Field	Value
Fixed	Rev 2 (anti-bluff review): minio dropped `:latest` for a placeholder pinned version tag (digest TBD) per the digest-pinning rule; k8s postgres probes now derive the user/db from `$POSTGRES_USER`/`$POSTGRES_DB` (Secret-sourced, not hardcoded `helix`) and the minio `mc helix` token is documented as an alias name, not a user; compose healthcheck `wget` flagged as Dockerfile-dependent (distroless/read-only bases); caddy `/healthz` flagged as dependent on the out-of-band Caddyfile (pending ADR-0004 QUIC spike); k8s `HELIX_S3_ENDPOINT=http://minio:9000` flagged as a placeholder that MUST be overridden (k8s omits in-cluster minio).
Continuation	Inspect the `containers`/`security`/`config` submodule surfaces and remove UNVERIFIED tags; confirm reverse-proxy choice (Caddy vs nginx vs `http3`-fronted Go) once the QUIC/UDP-443 spike (ADR-0004) closes; pin all image digests before any deploy; derive resource requests/limits and HPA thresholds from load tests; wire the dashboard build target.

Table of contents

> The table-of-contents requirement is mandated by HelixConstitution §11.4.61 (UNVERIFIED clause number). This document carries its ToC immediately after the metadata table.

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1. Purpose and scope

This document specifies **how the Helix OTA 1.0.0-MVP control plane is deployed**: the container set, the substrate it runs on, how the services connect, and how secrets are handled. It is normative for the MVP deployment topology and is the parent document for the concrete artifacts in this directory:

- `docker-compose.mvp.yml` — local / single-host topology.
- `kubernetes/` — orchestrated topology (namespace, ota-server Deployment+Service, postgres StatefulSet).
- `minio_setup.md` — object-store bucket/policy/key bootstrap.

Scope honors the locked decisions and does not re-decide them:

- **LOCKED stack (D6):** Go + Gin (gin-gonic) + Brotli + HTTP/3 (QUIC) via the `vasic-digital/http3` submodule with HTTP/2 + gzip fallback; **REST primary** (`/api/v1`); gRPC optional/internal only. [ADR-0004 §Decision; master §3]
- **LOCKED strategy:** native Android **A/B** on device (`update_engine` + AVB/dm-verity + automatic boot-failure rollback) + a **custom Go control plane** as a **modular monolith** for MVP (one deployable binary, extractable seams) per ADR-0003. [ADR-0003 §3]
- **LOCKED trust model (MVP):** per-artifact **signing + SHA-256 + AVB**; TUF device-side enforcement is **deferred to 1.0.1** per ADR-0002. The deployment must therefore carry signing/verification key material (see §5), not a TUF metadata server. [ADR-0002 §4.1, §4.3]
- **LOCKED state stores: PostgreSQL** relational state + **MinIO/S3** artifact blobs. [master §3; ADR-0001 §1; ADR-0003 §3]

Out of scope: the device-side agent deployment (it ships in firmware, not as a container), TUF/Uptane infrastructure (deferred to 1.0.1, ADR-0002), and any wrapped hawkBit deployable (gated on ADR-0001; if adopted it is a **separate** deployable behind ota-server, ADR-0003 §3).

2. Container substrate

All MVP services are **containerized and run on the `vasic-digital/containers` submodule substrate** (catalogue-first, §11.4.74 / §11.4.76, UNVERIFIED clause numbers). [master §10; submodule_reuse_map §3 Docs-export/substrate row] The `containers` submodule is the canonical catalogue name; **no new container-tooling submodule is invented** for this purpose.

- The **single ota-server deployable** (ADR-0003 modular monolith) is the only Helix-authored image; postgres, minio, the reverse proxy and the dashboard are off-the-shelf or build-from-`vasic-digital` images orchestrated by the same substrate.
- A future per-service split (ADR-0003 §3.2 scale trigger) **reuses the same substrate** — the topology files here change, not the substrate choice. [ADR-0003 §6 §11.4.76]
- Local/single-host orchestration uses Docker Compose (`docker-compose.mvp.yml`); orchestrated/multi-node uses the Kubernetes manifests (`kubernetes/`).

UNVERIFIED: the exact build/run interface the `containers` submodule exposes has not been inspected in this revision; the compose/k8s files here use standard OCI/Compose/Kubernetes schemas so they remain valid independently of that surface, and bind to `containers` for image build + substrate per the reuse map.

3. Service set

Service	Image (MVP)	Purpose	Exposed	Persists
ota-server	built from repo (`vasic-digital` Go build)	Go modular-monolith control plane (REST `/api/v1`, rollout engine, artifact intake, telemetry ingest)	internal `:8080` (HTTP/2 fallback) + `:8443/udp` (HTTP/3) behind proxy	no (stateless; state in postgres/minio)
postgres	`postgres:16`	Relational state (releases, deployments, devices, rollout phase/cohort state)	internal `:5432`	volume `pgdata`
minio	`minio/minio` (placeholder pinned version tag, never `:latest`; digest before deploy)	S3-compatible artifact blob store (`payload.bin` + manifests)	internal `:9000` (S3) + `:9001` (console)	volume `minio-data`
reverse proxy	`caddy:2` (placeholder, UNVERIFIED)	TLS termination, HTTP/3⍰HTTP/2 fronting, routing to ota-server + dashboard	public `:80`, `:443` (TCP+UDP)	no
dashboard	built from repo (placeholder, UNVERIFIED)	Operator web UI consuming ota-server REST	internal, via proxy `/`	no

3.1 ota-server (Go control plane)

- **Single deployable binary** per ADR-0003 (modular monolith with extractable artifact-validator / rollout-engine / OS-adapter seams). [ADR-0003 §3, §3.1]
- Serves **REST `/api/v1` as the primary surface** via Gin; HTTP/3 (QUIC) primary through the `vasic-digital/http3` drop-in `net/http.Handler` with

automatic HTTP/2 fallback; Brotli response compression negotiating to gzip. [ADR-0004 §Decision; master §3]
- **gRPC, if present, is internal-only** and never exposed through the public proxy. [ADR-0004 §Decision; additions_synthesis §5 C4]
- **Artifact path is NOT content-compressed** (the proxy and server must serve `payload.bin` byte-identical via `ZIP_STORED` + byte-range so device-side `FILE_HASH`/`METADATA_HASH` and the AOSP payload signature verify). This is a hard deployment rule. [ADR-0004 §Decision Option A on artifacts]
- Composes cross-cutting concerns from the catalogue: `http3`, `middleware` (Brotli/gzip), `ratelimiter`, `recovery`, `observability`, `config`, optional `cache`. [submodule_reuse_map §3 Transport/Caching/Config rows] (UNVERIFIED: each brick's exact surface.)
- Stateless: all state in postgres + minio, so it scales horizontally without sticky sessions.

3.2 postgres

- `postgres:16` (LTS-class major; pinned to a digest before deploy). Holds all relational state for the modular monolith. [ADR-0003 §3; master §3]
- A **separate schema/instance is recommended** if a wrapped hawkBit is ever adopted (ADR-0001), so they do not share a schema. [ADR-0001 §3.1; eclipse-hawkbit §6] Not needed at MVP since AOSP-native-only / Helix-only state is the baseline.
- Durable via a named volume (compose) / PVC (k8s StatefulSet).

3.3 minio (artifact blobs)

- S3-compatible object store for OTA artifacts (`payload.bin`, manifests, detached signatures). The `Storage` catalogue brick abstracts all access; **no direct S3 SDK calls live outside `Storage`**. [submodule_reuse_map §3 Storage row] (UNVERIFIED: `Storage` provides an S3/MinIO backend.)
- Bucket layout, access policy, and service-account keys are bootstrapped per `minio_setup.md`.
- Must support **range GET** for large artifacts (device-pull streaming); if `Storage` lacks range-get it is an upstream extend item (UNVERIFIED). [submodule_reuse_map §5; ADR-0004 §1 range-request caveat]

3.4 reverse proxy

- Terminates TLS (TLS 1.3 throughout, master §6), fronts **HTTP/3 (QUIC) on UDP/443 with HTTP/2 fallback on TCP/443**, and routes `/api/*` → ota-server, `/` → dashboard. [ADR-0004 §Decision; master §3, §6]
- Image is a **placeholder** (`caddy:2`, chosen for first-class HTTP/3) — **UNVERIFIED**; final choice depends on the QUIC/UDP-443 + Brotli + range-request spike flagged in ADR-0004 (the `http3` submodule may instead front the server directly, removing a separate proxy). [ADR-0004 §1, §Decision caveat]
- **Does not compress the artifact path** (§3.1 rule).

3.5 dashboard

- Operator web UI; a static/SPA build consuming ota-server REST `/api/v1`. Image built from the repo (**placeholder/UNVERIFIED** build target). Served behind the proxy.
- No business logic; it is a thin REST consumer. Auth flows reuse the `auth`/ `security` bricks via the server. [submodule_reuse_map §3 Auth row]

4. Topology and data flow

```
                       (public)
   devices  ──HTTP/3 QUIC :443/udp──┐
   operators ─HTTP/2 TLS :443/tcp──┐│
                                   ▼▼
                    ┌─────────────────┐
                    │  reverse proxy   │  TLS1.3, HTTP/3↔2,
routing
                    └───────┬─────────┘
                 /api/*     │     /  (dashboard)
                    ┌───────▼───────┐   ┌───────────┐
                    │   ota-server   │   │ dashboard │
                    │ (Go monolith)  │   └───────────┘
                    └───┬─────────┬─┘
              relational │         │ blobs (range GET, byte-
identical)
                    ┌───▼──┐   ┌───▼────┐
                    │ pg    │   │ minio  │
                    └──────┘   └────────┘
```

- **Control-plane traffic** (REST/JSON poll, rollout assignment, telemetry) is Brotli/gzip compressed. **Artifact traffic** (`payload.bin`) is served byte-identical, no compression, range-GET capable. [ADR-0004 §Decision]
- ota-server is stateless; postgres and minio are the only stateful services.

5. Secrets handling

Secrets are handled via the **`security` catalogue brick + container/orchestrator secret mechanisms** — **NOT** via any invented Vault submodule (no such submodule exists in the catalogue; inventing one violates §7.1 / §11.4.6). [submodule_reuse_map §2 canonical names; documentation_standards §9]

- **Crypto primitives & signing/verification keys** (per-artifact signing + SHA-256 verification, ADR-0002 MVP trust model) are owned by the **`security`** brick (server) / **`Security-KMP`** (device). The deployment's job is only to *supply* the key material to `security`, not to re-implement key handling. [ADR-0002 §4.1; submodule_reuse_map §3 Auth row] (UNVERIFIED: `security` exposes the signing/verify + offline-key primitives — flagged in ADR-0002 §8.)

- **Runtime secret delivery** uses the container substrate's native secret mechanism:
    - **Compose:** Docker `secrets:` (file-mounted under `/run/secrets/…`) and/or an env file excluded from VCS. The committed `docker-compose.mvp.yml` uses **env placeholders only** — no real credentials are committed.
    - **Kubernetes:** `Secret` objects mounted as files/env; the manifests reference Secret **names**, never literal values. (A sealed-secrets/external-secrets operator MAY back these in a real cluster — UNVERIFIED, deployment-environment choice.)
- **Secret classes at MVP:** (1) artifact **signing private key** (offline/HSM-preferred; only the *public* verification key needs to be online for server-side upload verification, ADR-0002 §4.1); (2) **postgres credentials**; (3) **minio access/secret keys**; (4) TLS certificate/key for the proxy; (5) OAuth2/JWT signing secrets owned by `auth/security`.
- **No secret value appears in any committed file.** The compose and k8s artifacts in this directory carry placeholders/Secret-name references only.

6. Environments: local (compose) vs orchestrated (kubernetes)

Concern	Compose (`docker-compose.mvp.yml`)	Kubernetes (`kubernetes/`)
Use	local dev, single host, demo	staging / production multi-node
ota-server	`build:` from repo	`Deployment` (replicas, rolling update) + `Service`
postgres	container + named volume	`StatefulSet` + PVC + headless `Service`
minio	container + named volume	(out of scope of MVP manifests; use operator/managed S3)
secrets	Docker `secrets` / env file	`Secret` objects (referenced by name)
scaling	manual	horizontal (Deployment replicas; HPA later, thresholds UNVERIFIED)

The MVP k8s set is intentionally minimal (namespace + ota-server Deployment/Service + postgres StatefulSet) to match the modular-monolith, lean-MVP bias (ADR-0003 §4.1). minio and the proxy are deferred to a managed/operator path in the orchestrated environment (UNVERIFIED choice).

7. Testing (four-layer)

Deployment artifacts are gated by the four-layer testing model (master §13; §1/§1.1 four-layer testing + mutation immunity, UNVERIFIED clause numbers). For *deployment* artifacts the four layers are:

1. **Source-presence gate (Layer 1).** Static existence/shape checks on the artifacts themselves: `overview.md`, `docker-compose.mvp.yml`, `kubernetes/namespace.yaml`, `kubernetes/ota-server-deployment.yaml`, `kubernetes/ota-server-service.yaml`, `kubernetes/postgres-statefulset.yaml`, and `minio_setup.md` all exist; each declares the required services/objects; **no committed file contains a literal secret value** (grep gate); every Markdown doc opens with the metadata table + ToC.
2. **Artifact gate (Layer 2).** The YAML is *valid and parseable*: `docker compose -f docker-compose.mvp.yml config -q` exits 0; `kubectl apply --dry-run=client -f kubernetes/` (or `kubeconform`) passes; healthchecks are present on every long-running compose service; image references are pinnable to digests. (This validates structure without running anything.)
3. **Runtime / integration gate (Layer 3).** Bring the stack up (`docker compose -f docker-compose.mvp.yml up`); assert each healthcheck reaches healthy; ota-server `/healthz` returns 200 over HTTP/2 fallback and HTTP/3; postgres accepts the server's connection; minio bucket from `minio_setup.md` is reachable and a round-trip artifact PUT/range-GET is **byte-identical** (artifact-path no-compression rule, ADR-0004); a signed test artifact passes server-side SHA-256 + signature verify (ADR-0002).
4. **Mutation meta-test (Layer 4).** The deployment tests must *fail when the deployment is wrong* — a meta-test that mutates the artifacts and asserts the gates catch it: e.g. remove a healthcheck → Layer 2 fails; point ota-server at a wrong DB host → Layer 3 fails; inject a plaintext secret into a committed file → Layer 1 grep gate fails; enable artifact-path compression → Layer 3 byte-identity check fails. A mutation that no gate catches is a missing test (mutation immunity). [master §13; §1.1]

Safety-critical paths exercised by deployment (signing/verify on upload, byte-identical artifact serving) inherit the **≥90% coverage floor**. [ADR-0003 §6 §1/§1.1; additions_synthesis §5 C8]

8. Open / UNVERIFIED items

1. **HelixConstitution clause numbers** (§11.4.6/7.1/§11.4.61/§11.4.74/§11.4.76/§1/§1.1) — carried from corpus convention; **UNVERIFIED**.
2. **Catalogue submodule surfaces** (`containers`, `security`, `config`, `observability`, `Storage`, `http3`, `middleware`) — exact public API not inspected this revision; "satisfies" claims **UNVERIFIED** (mirrors submodule_reuse_map §2/§3 and ADR-0002 §8).
3. **Reverse-proxy choice** (`caddy:2` placeholder) and whether the `http3` submodule fronts the server directly — **UNVERIFIED**; depends on the QUIC/UDP-443 + Brotli + range-request spike in ADR-0004 §1.
4. **Dashboard build target/image — placeholder/UNVERIFIED**.
5. **Resource sizing & HPA thresholds** (CPU/RAM/replicas; telemetry-vs-rollout split thresholds) — no load figures exist; set from MVP load tests. **UNVERIFIED.** [ADR-0003 §3.2, §7]

6. **Image digests** — `postgres:16`, `minio/minio`, `caddy:2` must be pinned to digests before any real deploy. **UNVERIFIED** until pinned.
7. **hawkBit adoption (ADR-0001)** would add a separate deployable (Spring Boot + its own postgres schema) behind ota-server; this topology holds either way. **UNVERIFIED.**

9. Sources

- `../../research/adr/adr-0001-wrapped-engine.md` — wrapped-engine decision; hawkBit-as-separate-deployable + separate-schema note.
- `../../research/adr/adr-0002-supply-chain-trust.md` — MVP trust model (signing + SHA-256 + AVB; TUF deferred to 1.0.1); `security` brick fit (UNVERIFIED).
- `../../research/adr/adr-0003-server-topology.md` — modular monolith (one deployable); `containers` substrate; scale trigger.
- `../../research/adr/adr-0004-transport.md` — HTTP/3→HTTP/2, Brotli/gzip, REST-primary, artifact-path no-compression rule.
- `../../00-master/2026-06-07-helix-ota-design.md` — master design (§3 transport, §6 security, §10 catalogue/containers, §13 testing).
- `../../00-master/submodule_reuse_map.md` — catalogue-first bindings; canonical submodule names; `Storage`/`security`/`http3` rows.
- `../../00-master/documentation_standards.md` — §9 canonical submodule catalogue (no invented names).